

BHUVANESH ARUN C T

+918838371488 | bhuvanesharun777@gmail.com | [linkedin.com/in/bhuvanesharunct](https://www.linkedin.com/in/bhuvanesharunct) | github.com/bhuvi-777 | [Portfolio](#)

Education

B.E -Electrical and Electronics Engineering
Chennai Institute of Technology

CGPA: 8.5
2023-2027

Technical Skills

Automation & Robotics: ROS2, KUKA KR16, KUKA LBR Issy 11, RoboDK, Fusion 360 | **Embedded Systems & IoT:** ESP32, Arduino, Node-RED, Sensor Integration, UART, SPI, I2C, REST/HTTP Dashboards | **PCB & Hardware Design:** KiCad, Altium Designer, PCB Routing, Gerber Generation, PCB Assembly, LTspice | **Programming & Simulation:** C, MATLAB/Simulink, HTML/CSS/JavaScript, C++ (Basic) | **Platforms & Tools:** Linux, Git/GitHub, Firebase, Industry 4.0, Smart Manufacturing, Continuous Improvement

Work Experience

Schneider Electric - Production Intern

Dec 2025 - Present | Bangalore, Karnataka

- Built 5 production dashboards and operational tools, reducing 120 manual man-hours and saving ₹20L in operational costs while adhering to production safety and quality protocols. Contributed to Node-RED-based IoT and Industry 4.0 initiatives, supporting data collection for process monitoring and decision-making. Currently leading integration of an Autonomous Mobile Robot (AMR) for smart manufacturing shopfloor automation.

PICABORD Technologies Pvt. Ltd. - Hardware Design Engineer

Apr 2025 -Dec 2025 | Chennai, Tamil Nadu

- Designed the complete schematic and PCB layout of a single-board computer from the ground up, covering component selection, signal routing, and hardware integration. Built hands-on proficiency in embedded system design and manufacturing-ready output.

Meera Laser Solutions Pvt. Ltd - Project Engineer Intern

Aug 2025 - Sep 2025 | Chennai, Tamil Nadu

- Led the setup and commissioning of a laser welding head on a KUKA robotic arm, managing end-to-end coordination across partner companies - including vendor alignment, equipment procurement, and delivery scheduling — to ensure seamless on-site execution.

Renom Wind Energy Services Pvt Ltd - Engineering Trainee

Nov 2024 - Dec 2024 | Tirunelveli, Tamil Nadu

- Completed a 6-week O&M programme on Enercon E40 wind turbines, performing field inspections and fault troubleshooting. Strengthened understanding of grid-connected electricity generation, turbine sub-systems, and preventive maintenance workflows.

Vels Wind Power - Lab Engineer Intern

May 2024 - Jun 2024 | Kanyakumari, Tamil Nadu

- Assembled and tested a 12V battery charger circuit PCB over a 45-day engagement, applying theoretical electronics knowledge in a live lab environment. Developed systematic circuit troubleshooting skills including error detection, root-cause analysis, and rectification.

Project

Six-Wheel Rocker-Bogie Autonomous Lunar Rover (Caterpillar Autonomy Challenge 2026)

Technologies: Arduino, Raspberry Pi, Motor Drivers, Sensor Integration, Autonomous Navigation

- Served as the Electrical & Electronics Lead for a six-wheel rocker-bogie autonomous lunar rover. Led component selection, electrical system integration and developed Arduino-based drivetrain control interfaced with Raspberry Pi for autonomous navigation, deployed in competition under Caterpillar Autonomy Challenge 2026.

Custom Arduino Nano PCB Design using KiCad Designer

Technologies: KiCad, Gerber generation

- Designed a custom Arduino Nano-compatible PCB using KiCad, executing the complete hardware design flow - schematic capture, optimized PCB routing, 3D visualization, and Gerber file generation.

Power System Load Flow Simulator (Full-Stack Web Application)

Technologies: React, Node.js, Express, React Flow, Zustand, Vitest, GitHub Pages, Render.

- Built an interactive, browser-based load flow simulator featuring drag-and-drop network modelling, automated Y-bus matrix generation, Gauss-Seidel solver, convergence visualization, and PDF export. Implemented the application using React, Node.js, and Express, deploying the frontend on GitHub Pages and backend APIs on Render.

IoT-Based Smart Protection System for Electrical Appliances with Predictive Anomaly Detection

Technologies: ESP32, DS18B20 sensor, Current/Voltage sensor, SSR/Relay, Web Dashboard (HTTP), Firebase, Machine Learning

- Developed an IoT-enabled appliance safety system using ESP32 for real-time temperature, current, and voltage monitoring. Integrated automatic power disconnection on fault detection, remote web dashboard monitoring, instant alert notifications, and predictive anomaly detection to proactively prevent overheating and electrical hazards.

Research Activity

Authored and submitted a research paper on power quality challenges from high-penetration EV integration and grid adaptation strategies to *Energy Exploration & Exploitation* journal.

Certification

Universal Robotics | Introduction Energy Literacy Training | Cisco Cybersecurity Essentials & Introduction | Altium PCB Design | NPTEL Introduction to IoT | NPTEL Digital Circuits | AICTE Embedded System Developer Virtual Internship

Achievements

Finalist – Caterpillar Autonomy challenge 2026 , IIT- Madras | Winner – Roborace event, Takshashila 2025, CIT Chennai | Finalist – Autorace (RC Racing) event, Kurukshetra 2025, CEG Chennai | **Finalist – Deepsprint Hackathon (Top 30 out of 800 teams), CIT Innovation Labs, Innovest 2025** | Finalist – Nexathon 2024, CIT Chennai

Extra-Curricular Activities

- Attached to KUKA's Centre of Excellence; hands-on experience operating KUKA KR 16 and KUKA LBR Issy 11 robots.
- Organised the "Automation and Robotics Workshop" by KUKA Training Centre at Innovest 2024.
- Served as Joint Secretary and Designer for Impulse 2025 (Symposium), overseeing event branding and visual communications using Adobe Photoshop and other design tools.